

Problem Set 2

Directions

Submit two files on Moodle before the deadline:

- [1] your **solution set** as a *.pdf [2] the **R script** as a *.R type file

For both, the first line should include your student ID and the assignment number

If you worked with one or two other students write "group:" and then their names

[1] the **solution set** should include for each answer:

- the question number/part
- your answer in words
- any hand drawn diagrams
- the command in R that was used (if relevant)
- the output from Rstudio (if relevant)

[2] the **R script** should include:

- all the commands you used to generate the solutions
- individually written comments that explain your code to the grader

Women's fertility and education level

We are interested in studying the effect of education on women's fertility (number of children). We have pooled cross-sectional data from the National Opinion Resource Center's General Social Survey. A similar dataset was used by Sander (1990) to study the effect of women's education on the number of children they have. For each woman in the sample, we have the following:

- **year**: the year the survey was taken
- **childs**: total number of children the woman has (1 – 7, or NA if 0)
- **age**: the age of the woman (18 – 88)
- **educ**: the number of years of education the woman has
- **race.Black**: indicator if the woman's race is Black
- Type of place the woman lived in at age 16:
 - **bhcity**: indicator for suburb of a big city, or a big city
 - **town**: indicator for a town or small city
 - **farm**: indicator for a farm or other open country
- Region of the USA the woman lived in at age 16:
 - **west**: indicator for the Pacific or Mountain regions of the USA
 - **east**: indicator for the North East or Mid-Atlantic regions of the USA
 - **south**: indicator for the Southern regions of the USA
 - **central**: indicator for the Midwest and Central regions of the USA

Note that there are two categorical variables: type of place the woman lived in at age 16 (big city, town, or rural), and region of the US the woman lived in at age 16 (West, East, South or Central).

1. Read the discussion below from Sander (1990). Why might education have a causal effect on the number of children a woman has? Why is the variable education endogenous with number of children? Use your own words to summarize the discussion in a few sentences.

1. Introduction

Numerous studies indicate that women's schooling has a negative effect on fertility. Some of the possible explanations for this effect include (1) women's schooling increasing the value of time in market work and, accordingly, the opportunity cost of allocating time to child rearing [Becker (1991)]; (2) women's schooling increasing the efficiency of fertility control [Michael (1973)]; and (3) schooling reducing preferences for children [Cleland and Wilson (1987), Easterlin (1989)].

Women's schooling could also be endogenous with fertility since children affect the cost of schooling for mothers. Mothers with children at home relative to other women are less likely to make additional investments in schooling. Perhaps more importantly, women's schooling could be correlated with unobservable traits that are jointly determined with fertility. Boulier and Rosenzweig (1984) found this to be the case in a study of women in the Philippines; they found that the negative correlation between schooling and fertility was a product of selection. In a related study, Sander (1992) found that schooling is endogenous with marital status. Thus, schooling should not be treated as an exogenous determinant of decisions that are made in the marriage market. Other studies also indicate that schooling and various types of behavior might be jointly determined [see, for example, Farrell and Fuchs (1982) and Fuchs (1982)].

2. Draw a causal diagram showing the relationship between the following variables: education, age, race, number of children, type of place at 16, region of USA at 16. The variable education should be endogenous with number of children. You should also include "*unobserved cultural norms at 16*" somewhere in your diagram.
3. Import the data into R and create a subset of the data for women aged 35 – 54. Using this subset run an OLS regression which estimates the relationship between education and number of children, controlling for any variables mentioned in question 2 that you think are appropriate. Explain why you chose the control variables you chose and why you omitted the variables you omitted. Discuss the effects of your control variables on fertility.
4. Estimate your same model you used in (3) but this time use **paeduc** (father's years of education) and **maeduc** (mother's years of education) as instruments for the woman's education. Interpret your result for the coefficient on education.
5. Why is the coefficient on education different in your results from (3) vs (4). What is each one capturing?
6. Are mother's education level and father's education level relevant instruments?
7. Do you think mother's education level and father's education level are valid instruments for education level? Explain your reasoning.
8. You suspect there may be endogenous selection into your sample. Specifically, you suspect that the year a woman was born (**year.born**) affects whether or not she has children. Create a variable that indicates if a woman has at least 1 child. Using the same model as in (3), run Heckman correction for selection. Is there evidence of selection into the sample?